

# A novel solution to manufacturing multi-color medical preoperative models with transparent shells

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## ABSTRACT

Preoperative models manufactured by 3D printing technology are widely demanded in surgical procedures. Current printing methods such as Polyjet, color jet printing (CJP) and fused deposition molding (FDM) are either too expensive or of poor performance. In this paper, a novel solution is proposed. Preoperative models are first divided into internal part, such as blood vessel, and the main body. The former is printed in color and the latter is transparent. In particular, the internal part is printed using FDM. A housing template for the main body is generated by digital light processing (DLP) based on the preoperative model. After integrating the template and inner part, transparent elastomer precursor is injected and cured inside the template, leading to the formation of the transparent main body. For demonstration, a liver model is manufactured using the proposed scheme. Compared with the existing methods, our strategy could manufacture preoperative models at both high quality and affordable cost.

## CCS CONCEPTS

• **Computing methodologies** → *Shape modeling*.

## KEYWORDS

Medical preoperative model, 3D printing, Model reconstruction

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## 1 INTRODUCTION

Preoperative models, namely "cloned" 1:1 pathological models, are widely demanded in surgical procedures. They can help doctors to develop reasonable surgical plans and improve the success rate of surgery. Moreover, they also enable patients and their families

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**Table 1: Comparison of different preoperative model manufacturing methods**

|         | Manufacturing cost | Surface roughness | Color fidelity | Transparent or not | Easily damaged |
|---------|--------------------|-------------------|----------------|--------------------|----------------|
| Polyjet | High               | Low               | High           | Yes                | No             |
| CJP     | High               | High              | High           | No                 | Yes            |
| FDM     | Low                | High              | Low            | No                 | No             |
| Ours    | Middle             | Low               | High           | Yes                | No             |

to understand the condition more intuitively. 3D printing is the most suitable technology for the fabrication of preoperative models because of its great shaping ability, designing flexibility and personalized manufacturing.

Current 3D printing technologies for preoperative model manufacturing include Polyjet, CJP and FDM (Table 1) [Chen et al. 2020; Wei et al. 2017]. Polyjet can print products with high color fidelity, low surface roughness, and better robustness, but the printing cost is high. CJP technology can make products with high color fidelity, but cannot print transparent items. In addition, the printed products are fragile and have high surface roughness. The printing cost of CJP is similar as Polyjet. FDM is very cheap and the printed models are robust. However, the surface roughness of printed products is high, and the color fidelity is low. Furthermore, it is difficult to print transparent parts using FDM.

Obviously, none of the above techniques can simultaneously satisfy all the demands in manufacturing preoperative models. Therefore, better 3D printing solutions are highly desirable. In this paper, we propose a novel method to manufacturing multi-component medical preoperative models with transparent body. The model is divided into internal part such as blood vessel and the main body. The internal part is printed in color using a FDM printer [Wu et al. 2021]. A housing template is generated by DLP based on the preoperative model [Wu et al. 2018]. Then the printed housing template and the inner part are integrated. A transparent elastomer precursor is injected and cured inside the template, leading to the formation of the transparent main body. Based on this proposed strategy, a liver model is manufactured. Compared with existing methods, our technique could manufacture preoperative models at high quality, high color fidelity and affordable cost (Table1).

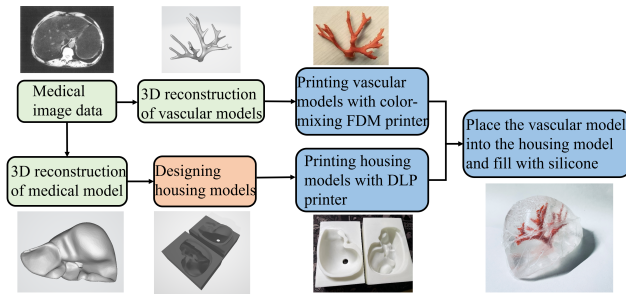


Figure 1: Framework of the proposed method.

## 2 FRAMEWORK OF THE PROPOSED METHOD

Figure 1 shows the framework of the proposed scheme. Using CT data as inputs, image analysis is firstly performed to segment the vascular region and organ contour region, and then the Marching Cube (MC) algorithm is used to obtain the 3D model of blood vessels and organs. Secondly, the housing template is generated from the 3D organ model. Next, the vascular model is printed using color-mixing FDM. The housing template is printed using a DLP 3D printer. Finally, the housing template and the vascular model are assembled, and an elastomer precursor is infused into the template and solidified. After removing the housing template, the manufacturing process is finished.

## 3 DESIGNING THE HOUSING TEMPLATE

First, the small part of blood vessels outside the liver model (boxed region in Figure 2(a)) will be mounted onto the housing template, so that the blood vessels can be fixed. To facilitate the fixation, the direction of the vessels should be downward.

Next, we obtain the bounding box of the preoperative model in the printing direction, and extend the length, width, and height by 20 mm to obtain a rectangular box larger than the actual object. We subtract the model from the rectangular box to obtain the template with a hollow organ outline.

In order to insert the vascular model, the housing template is divided into multiple parts. For the liver model in this paper, the housing template is divided into two parts,  $Part_{Up}$  and  $Part_{Bottom}$ . In order to fix the vascular model, it is necessary to ensure that the vascular embedding slot can be completely inserted into  $Part_{Bottom}$ . Therefore, the location with the largest cross-sectional area above the exposed vessels is used as the boundary between  $Part_{Up}$  and  $Part_{Bottom}$ . In order to prevent elastomer leakage,  $Part_{Up}$  and  $Part_{Bottom}$  are fixed by interlocking designs (Figure 2(b)). A filling hole is added at the top of  $Part_{Up}$ .

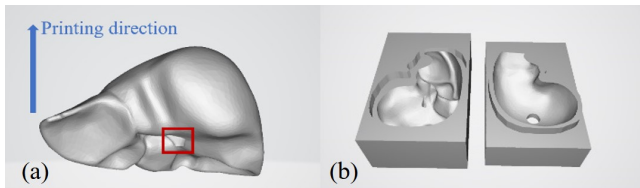


Figure 2: (a): Liver model. (b): The housing model.

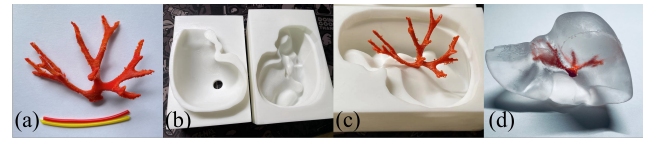


Figure 3: Illustration of the manufacturing process.

## 4 PRINTING SAMPLES

The color mixing FDM 3D printer is equipped with an extrusion head with 3-input and 1-output. It allows 3 differently colored filament to be mixed. Various colors could be obtained by varying the feeding rate of the three filaments [Wu et al. 2021]. Figure 3(a) shows the printed blood vessel model, which was printed by mixing red and yellow filaments at a ratio of 8:2. The housing template (Figure 3(b)) was printed using a DLP 3D printer [Wu et al. 2018].

The vascular model, PartBottom and PartUp are assembled as Figure 3(c). The silica gel, Sylgard 184, is infilled into the integrated housing template and kept at RT for 24 hours to complete solidification. The housing template is removed and the complete organ could be obtained as shown in Figure 3(d)

From Figure 3, our method can manufacture the transparent model. And the color fidelity is high. The surface roughness is low. Based on the property of silica gel, Sylgard 184, the manufactured model is not easy damaged. Furthermore, the material cost is compared. The material cost of Polyjet, CJP and FDM are \$0.5/g, \$0.43/g and \$0.04/g respectively, while the material cost of the proposed method is \$0.16/g, which is much lower than Polyjet and CJP. Therefore, the proposed scheme can produce high-quality preoperative models with the middle cost.

## 5 CONCLUSION

We propose a low-cost preoperative model manufacturing method that combines color-mixing FDM and DLP 3D printing technology, and the preoperative liver model was successfully made. In addition to liver models, more preoperative models with complex geometric structures will be analyzed and manufactured in the future.

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